

8. EXPERIENCE: 30 years (UG) and 9 years (PG) teaching experience and 24 years Research experience.

9. AREA OF SPECIALIZATION : Solid State Physics, Optoelectronics, Sensors and Transducers, Polymers, etc

10. Research Publications in International Journals:

1. Vinod S. More, M. P. Patil, R. P. Tandel, **T.N. Ghorude**, Gitesh G. Padhye published research paper entitled, **“Study of Structural, Morphological, Electrical and Thermal Characteristics of PPy-Fe(ClO₄)₂-Au Nanocomposite and its Application in Ammonia Gas Sensing”** in *Indian Journal of Science and Technology(IJST)* (**UGC Care listed Journal**). Vol-17, Issue-46, 18th December 2024, Page No. 4842-4853.
2. Raju P. Tandel, Vinod S. More, **T.N. Ghorude** has published paper entitled **“INVESTIGATING THE ROLE OF FE ON STRUCTURAL, MORPHOLOGICALAND ELECTRICAL CHARACTERISTICS OF POLYPYRROLE NANOPARTICLES”** in *International Journal of Scientific Research*. ISSN: 2277-8179, Volume-13, Issue-08, August-2024, pp 01-03, Impact Factor 5.711
3. Vinod S. More, **T.N. Ghorude**, Raju Tandel, B.K. Sakhare, G.G. Padhye has presented paper entitled **“From Iron to Sensing: Investing the Role of Iron Chloride in Polypyrrole based LPG Gas detection”** in one day National conference on **“Recent Developments In Basic And Applied Sciences (RDBAS 2024)”** organized by Dr.S.D.D. Arts, commerce and Science college, Wada, Palghar on Saturday, 2nd March 2024. Also, Published in **Research Journey**, International E- Research Journal, E-ISSN 2348-7143, Impact Factor – 6.625, **Special Issue 338 : Recent Developments in Basic and Applied Sciences**, March 2024, page No. 194-202.
4. Vinod S. More, **T.N. Ghorude**, Apurva M. Patil has **presented** paper entitled **“Synthesis of Polypyrrole Nanoparticles for enhanced H₂S Gas Sensor Fabrication”** in one day **International Multidisciplinary e-conference on Recent Trends in Science, Commerce & Humanities (IMCRTSCH-2024)”** organized by D.V. College, Goregaon, Raigad on Saturday, 16th March 2024. **Published in International Multidisciplinary E-Journal**. ISSN 2277-4262, **Impact Factor – 8.285**, Vol-13, Issue-05, 01st May 2024, page No. 237-246.
5. Minal P. Patil, Gaurav Singh, Vinod S. More, **T.N. Ghorude**, **“DEVELOPMENT OF MICROCONTROLLER BASED HAEMOGLOBIN METER”** published in

International Journal of Scientific Research. ISSN: 2277-8179, Volume-13, Issue-02, February 2024, pp 47-49, Impact Factor 5.711

6. B. K. Sakhare, **T.N. Ghorude**, V.S. More, **“Polypyrrole Infused Cotton Fabrics: An Elegant Solution for Gas Sensing”** published in **International Journal of Scientific Research (IJSR)** ISSN: 2277-8179 Volume-12 Issue-11, November 2023 page No.66-68, Impact factor 5.711
7. Sahebrao B. Pagar , **Tatyrao N. Ghorude** , Maria P. Nikolova , K. SenthilKannan **“Synthesis, physical, chemical, biological, mechanical and electronic studies of polypyrrole (PPy) of versatile scales for electro-mechano, pharmaceutical utilities”** Science Direct **Heliyon** Volume-9, Issue-9, <https://doi.org/10.1016/j.heliyon.2023.e20086> Received 17 May 2023; Received in revised form 24 August 2023; Accepted 11 September 2023 Available online 12 September 2023 Impact factor 3.776
8. Sahebrao. B. Pagar, Dr. Minal. P. Patil, **Dr. Tatyrao. N. Ghorude** **“quantative color measurement of ph indicator papers using developed tristimulus colorimeter and hr-4000 high resolution spectrophotometer”** [International journal of scientific research](#) Volume - 12 | Issue - 09 September - 2023 | PRINT ISSN No. 2277 - 8179 | DOI : 10.36106/ijsr Impact Factor 5.711
9. Sahebrao B. Pagar, **Tatyrao N. Ghorude**, Mrinalini D. Deshpande, K. SenthilKannan **“Facile Fabrication of Room Temperature Based H₂S Gas Sensor Using ZTO-Ag@PPy Hybrid Nanocomposite”** **Springer** Journal of Inorganic and Organometallic Polymers and Materials <https://doi.org/10.1007/s10904-023-02650-8> published online: 24 May 2023 Impact factor 3.518
10. Vinod S. More , **T. N. Ghorude** , Supriya N. Save , B. K. Sakhare , R. P. Tandel **“Quantification of Fe (II) Ions in the Synthesis of Polypyrrole by Spectrophotometric Detection”** International Journal of Science and Research (IJSR) ISSN: 2319-7064 Volume 12 Issue 4, April 2023 page No. 1812 – 1816 Impact factor 7.942
11. **Dr. T.N.Ghorude** ,N. S. Shah, V.D.Thakur **“MICROCONTROLLER BASED FLAT-FOOT DETECTOR USING LOAD CELLS”** INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH , Volume - 11 | Issue - 09 | September - 2022 | PRINT ISSN No. 2277 - 8179 | DOI : 10.36106/ijsr Page No. 13-14. Impact Factor 5.862
12. R. Mundotia , **T. Ghorude** , D.C. Kothari c , A. Kale b , N. Thorat **“Tuning bilayer period of AlN/CrN super lattice coatings deposited using cathodic arc technique for superior mechanical properties and thermal stability”** ScienceDirect **ELSEVIER** Applied Surface Science Advances Volume 7, February 2022, Impact Factor-6.2

13. **Dr. T.N.Ghorude**, G. G. Singh “ **μ C based Fuel quantity and density measurement**” International Journal of All Research Education and Scientific Methods (IJARESM), ISSN: 2455-6211 Volume 9, Issue 10, PP 351-354 October -2021, Impact Factor: 7.429,
14. **T. N.Ghorude** and M. P. Patil “**SPECTRAL RESPONCES OF C.I.E.1964 R.G.B. DATA USING PYTHON**” International Journal of Advance and Innovative Research ISSN: 2278-7844, Volume 8, Issue 2 (III), April - June 2021. Impact Factor-7.15
15. R. Mundotia, **T. Ghorude**, A. Kale and U. Mhatre. “**Study of multilayer and multi-component coatings deposited using cathodic Arc technique on H-13 hot work steel for die-casting applications**” **International Journal of Nano Dimension**, Volume 11, Issue 2, April 2020, Pages 177-187
16. R. Mundotia, D.C. Kothari, A. Kale, U. Mhatre, K. Date, N. Thorat, **T. Ghorude**. ‘ **Effect of ion bombardment and micro-blasting on the wear resistance properties of hard TiN coatings**’ **ELSEVIER Materials Today: Proceedings** Volume 26, Part 2, 2020, Pages 603-612. Impact Factor-31.041
17. R. Mundotia, N. Thorat, A. Kale, U. Mhatre, D. C. Kothari, T. Kovacs and **T. Ghorude**. “**Study of corrosion properties of CrN and multilayer CrN/Cr coating at different electrolyte temperatures deposited on stainless steel by vacuum arc process**” **AIP Conference Proceedings** Volume-2115, Issue-I, July-2019.
18. M.P.Patil and **T.N.Ghorude** “**pH Indicator paper Color measurement using wireless tristimulus colorimeter**” International Journal of Scientific Research Volume-7, Issue-7, July-2018, ISSN 2277 – 8179, IF: 4.758, IC Value: 93.98 Pages 525-526.
19. M. P. Patil and **T. N.Ghorude** “**Measurement of hemoglobin (hb) in blood using photo detector by reflection method**” International Journal of Scientific Research Volume-6, Issue-9, September-2017, ISSN No 2277 - 8179, IF : 4.176 | IC Value : 78.46 Page No. 583-584.
20. **T.N.Ghorude** and M.P.Patil “**CIE standard 23 shades Color Measurement Using Wireless Tristimulus Colorimeter**” **International Journal of Chemical and Physical Sciences**, ISSN:2319-6602 IJCPS Vol. 5 Special Issue Nov. 2016 pp 15-20.
21. **M.P.Patil** and **T.N.Ghorude** “**Identification of Biological colonies using wireless digital colorimeter**” International Journal of Advances in Management Technology and Engineering Sciences Volume-6, Issue 2(1), March 2016 PP 18 – 21.
22. Mrs. Chinmaya Bari and **Dr. T.N.Ghorude** “**FACE RECOGNITION USING SKIN COLOR SEGMENTATION OF YCBCR AND RGB COLOR MODELS**” International Journal of Emerging Technology in Computer Science & Electronics (IJETCSE) ISSN: 0976-1353 Volume 23 Issue 1 –JUNE 2016. Impact Factor-4.78

23. **T. N. Ghorude**, D. B. Gaikwad and A. D. Shaligram “**Microcontroller Based Color Measurement Using RGB LEDs**” International Journal of Scientific Research ISSN-2277-8179 Vol. 1 issue 3 Aug 2012, Page No. 93-95. Impact Factor-5.711
24. **T.N.Ghorude**¹, Apurva M. Patil², Urmila L. Chimada³ “**Development of Hazardous gas detection system using Arduino R3**” International Journal of All Research Education and Scientific Methods (IJARESM), ISSN: 2455-6211, Volume 9, Issue 10, October -2021, Impact Factor: 7.429

International conferences

25. Vinod S. More¹, **T. N. Ghorude** has presented paper entitled “**From Particles to Fibers: Harnessing Polypyrrole's Conductivity for Gas Sensing Applications**” in one day International Conference on **Recent Advances in Material Sciences (RAMS-2024)** organized by ASP college, Devrukh
26. Vinod S. More, **T.N. Ghorude** “**Study of Structural and electrical properties of surface modified PPy-AuNPs thin film for gas sensing application**” presented in International Conference on ‘Advanced Materials for Physical, Chemical and Biological Applications’ funded by ‘Rashtriya Uchchatar Shiksha Abhiyan,’ jointly organised by the K.B.P. College, Vashi on 03rd & 04th March 2023.
27. Bhimrao Sakhare, R.P. Tandel, **T.N. Ghorude**, G.G. Padhye “**Synthesis and Characterization of Polypyrrole composites using various dopants**” presented in International Conference on ‘Advanced Materials for Physical, Chemical and Biological Applications’ funded by ‘Rashtriya Uchchatar Shiksha Abhiyan,’ jointly organised by the K.B.P. College, Vashi on 03rd & 04th March 2023.
28. Vinod S. More, Bhimrao Sakhare, Raju Tandel, Gitesh G. Padhye, **T. N. Ghorude** “**Smart Conducting Fabrics: Applications in Bio-Medical Sciences**” presented in International Conference on Recent Innovations in Science and Technology-2023 (RIST-2023) at I.Y. college between 10th and 11th Jan 2023
29. Bhimrao Sakhare, Raju Tandel, Vinod S. More, Gitesh G. Padhye, **T. N. Ghorude**, “**Synthesis of Polypyrrole based Smart Conducting Fabrics: Applications in Bio-Medical Sciences**” presented in International Conference on Recent Innovations in Science and Technology-2023 (RIST-2023) at I.Y. college between 10th and 11th Jan 2023
30. Raju Tandel, Bhimrao Sakhare, Vinod S. More, Gitesh G. Padhye, **T. N. Ghorude**, “**Electrical Properties of Polypyrrole Conducting Polymer at Various Dopant Concentrations**” presented in International Conference on Recent Innovations in Science and Technology-2023 (RIST-2023) at I.Y. college between 10th and 11th Jan 2023

31. **T.N.Ghorude¹**, Apurva M. Patil², Urmila L. Chimada³ **“Development of Hazardous gas detection system using Arduino R3”** Presented in International Conference on Advanced Nanomaterials & Nanotechnology jointly organised by Sonopant Dandekar College Palghar and Dept of Physics University of Mumbai between 26th to 28th Nov 2019.
32. Rajesh Mundotia^{1,2}, Ashwin Kale², Umesh. Mhatre², **T. N. Ghorude¹** **“Study of Multilayer and Multi-Component Coatings Deposited Using Cathodic Arc Technique on H-13 hot work steel for Die- Casting Applications”** Presented in International Conference on Advanced Nanomaterials & Nanotechnology at Sonopant Dandekar College between 26th to 28th Nov 2019.
33. **T. N.Ghorude¹** and M. P. Patil² **“SPECTRAL RESPONSES OF C.I.E.1964 R.G.B. DATA USING PYTHON”** Presented in International Conference on Advanced Nanomaterials & Nanotechnology jointly organised by Sonopant Dandekar College Palghar and Dept of Physics University of Mumbai between 26th to 28th Nov 2019.
34. **T.N.Ghorude¹** G. G. Singh² Kapil Wag^{h3} **“ μ C based Fuel quantity and density measurement”** Presented in International Conference on Advanced Nanomaterials & Nanotechnology jointly organised by Sonopant Dandekar College Palghar and Dept of Physics University of Mumbai between 26th to 28th Nov 2019.
35. RAJESH MUNDOTIA ,Others & **Dr.T.N.Ghorude.** **“Effect of ion bombardment and micro-blasting on the surface profile of TiN coated HSS samples deposited by Cathodic Arc Physical Vapour deposition (CA-PVD) technique”** in International Conference on Thin Film, November 13 to 17. 2017. at CSIR
36. **M.P.Patil** and T.N.Ghorude **“Identification of Biological colonies using wireless digital colorimeter”** international multidisciplinary conference at G.S.P College Talasari in association with **INAAR** on 26th March 2016.
37. **T. N.Ghorude** and M.P.Patil **“CIE standard 23 shades Color Measurement Using Wireless Tristimulus Colorimeter”** International Conference On Photon: Multiple & Creative Solution to Challenges (ICPMCSC) Organized by Dept. Of Physics Ratnam college, Mumbai 4th and 5th Dec.2015.
38. **T. N. Ghorude** and A. D. Shaligram **“Wireless online Color measurement using TCS 230 Color Sensor”** 3rd International Science Congress (ISC-2013) Conference at **Karunya University, Coimbatore**, Tamil Nadu, India from 8th - 9th December 2013.
39. **T. N. Ghorude**, A. L. Chaudhari and A. D. Shaligram **“Quantitative Color Measurement of pH Indicator paper Using Trichromatic LEDs and TCS230 Color Sensor”** International conference on Experimental Mechanics (ICEM08) Organized by The Chinese

Society of Theoretical and Applied Mechanics (CSTAM), **Southeast University, Nanjing, China (SEU) during 08th – 11th Nov. 2008 at Nanjing, CHINA - Presented Orally**

40. **T. N. Ghorude** and A. D. Shaligram “**Skin Color measurement using Tristimulus LEDs and TCS 230 Color sensor**” International conference on Microwaves and Optoelectronics (ICMO2007) Organized by Dr. B.A.M. University, Aurangabad, December 17-20, 2007.

Research Publications in National Journals/conferences

41. Vinod S. More, **T. N. Ghorude** has presented paper entitled “**Development of Low ppm Ammonia Gas Sensor using Polypyrrole-Gold Nanoparticles**” in one day **National Conference on “Sustainable Development in Society, Economy and Science”** organized by St. John College, Palghar on Saturday, 4th May 2024.
42. Vinod S. More, B.K. Sakhare, R.P. Tandel, G.G. Padhye, **T.N. Ghorude**, “**Electric Elegance: Enhancing Cotton Fabric with Conducting Polypyrrole for Gas Sensing Applications**” accepted for publication in **Mapana Journal of Sciences**, comes under **UGC Care listed journals**
43. **B.K. Sakhare**, V.S. More, T.N. Ghorude, “**Spectrophotometric Approach to Quantify the Iron(II) Ions in the Synthesis of Polypyrrole**” published in **Cosmos Multidisciplinary Research E-Journal**. ISSN: 2456-1665 Volume 8 Issue 10, October 2023 page No. 09-15, Impact Factor 5.3.
44. Vinod. S. More, **T.N. Ghorude**, M.P. Patil, B.K. Sakhare, R.P. Tandel, G.G. Padhye, “**Comprehensive Analysis of γ -radiation Impact on PVA/PTH Composite Films for Enhanced Battery Functionality**” published in **Indian Journal of Science and Technology**. ISSN: 0974-6846 Volume 16 Issue 46, 13 December 2023, page No. 4349-4357.
45. **T. N. Ghorude** and A. D. Shaligram “**Microcontroller driven RGB LED system for tristimulus surface colorimetry**” Indian journal of Applied Research, ISSN: 2249-555X, Volume-1, Issue-5, February 2012, Page No.81-83.
46. Chinmaya Bari and **T. N. Ghorude** “**Artificial Neural Network Based Pattern Recognition System**” National Conference on Emerging Trends in Physical Sciences (NCETPS) Organized by Dept. Of Physics N.B.Mehta science college, BORDI 23rd and 24th Jan 2015.
47. Minal P.Patil and **T. N. Ghorude** “**Measurement of chromaticity coordinates of pH Indicator paper using RGB colorimeter and software techniques**” National Conference on Emerging Trends in Physical Sciences (NCETPS) Organized by Dept. of Physics N.B.Mehta science college, BORDI 23rd and 24th Jan 2015.

48. **T. N. Ghorude** and A. D. Shaligram “**Wireless Tristimulus Colorimetry for Skin Color Measurement**” National Conference on Emerging Trends in Physical Sciences (NCETPS) Organized by Dept. of Physics N. B. Mehta science college, BORDI 23rd and 24th Jan 2015
49. **T. N. Ghorude** and A. D. Shaligram “**Microcontroller Based Color Measurement Using RGB LED’s and TCS230 Colour Sensor**” Deccan Current Science Vol. 6 No. 1 January 2012 pp 395-398.
50. **T. N. Ghorude**, and A. D. Shaligram “**Microcontroller driven RGB LED system for tristimulus surface colorimetry**” 13th National seminar on Physics and Technology of Sensors (SPTS-13) organized by Department of Electronic science, Pune university, Pune, 3rd March to 05th March 2008.
51. **T. N. Ghorude**, Jinesh P. Shah and A. D. Shaligram “**Color Recognition Using Artificial Neural Network**” National Conference on Physics of Semiconductor Devices and Smart Materials Organized by Karmaveer Bhaurao Patil College Vashi, Navi Mumbai, 22-23 December, 2007
52. **T. N. Ghorude** and A. D. Shaligram “**Color recognition by using RGB LED cluster and software techniques**”, 12th National seminar on Physics and Technology of Sensors (SPTS-12) organized by **Bhabha Atomic Research Centre, Mumbai** 400085, 07th to 09th March 2007
53. **T. N. Ghorude** D. B. Gaikwad and A. D. Shaligram “**Microcontroller Based Color Measurement Using RGB LEDs**”, 11th National seminar on Physics and Technology of Sensors (SPTS-11) organized by Centre for Sensor studies and Dept. of Electronic Science University of PUNE- 411007, 27th Feb to 01st March 2006
54. **T. N. Ghorude**, D. B. Gaikwad and A. D. Shaligram “**Optical fiber Based color measurement by using RGB LED’s**”, National seminar on Interdisciplinary Applications of Electronics organized by Arts, science & commerce college Chopda, Dist. Jalgaon 28 – 29th, January 2006
55. **T. N. Ghorude** and A. D. Shaligram “**The CIE color matching functions & calculations of CIE chromaticity co-ordinates under C++**”, National level workshop on Device-grade Material Development Using Ion Beams, jointly Organized by Nuclear science centre, New Delhi and Department of Physics, University of Mumbai during 3rd and 4th November 2003.